

SILIN RAJ GUPTA

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Education

University of Waterloo

Bachelor of Computer Science, Honours, Co-op

Waterloo, ON

Sep 2024 – Apr 2029 (Expected)

Professional Experience

Applied ML Engineer Intern

May 2026 – Aug 2026

Shopify — Search Product & Tooling

Toronto, ON

- Built the evaluation system for Shopify's autonomous ML experimentation agent, using **75 test cases and 10 scoring rubrics** to catch regressions and compare changes before they were merged.
- Moved Shopify's Python ML-pipeline tooling into the public TangleML/tangle-cli repository, preserving identical behavior while removing **5k+ lines of duplicated internal code**.
- Fixed a security issue where an AI agent could expose API tokens in generated pipelines, adding safe secret handling and pre-submission checks, validated across **3 end-to-end runs with zero secret-access attempts**.
- Benchmarked Shopify's agent against four alternative approaches across 110 experiments and 22 ML competitions; it earned **17/22 medals and 12 golds**, the highest gold-medal count.

Member of Technical Staff

Jan 2026 – Apr 2026

E3 Group

San Francisco, CA

- Rebuilt a production AI system that turns freight invoices into structured billing data, improving extraction accuracy from **70% to 98%+** through better prompting, validation, and customer-specific processing.
- Took ownership of production issues across the invoice platform, diagnosing live failures and duplicate billing and adding automated validation, recovery, and delivery safeguards to prevent recurring issues.
- Created a **shared React component library** for E3's frontends and used it to build and redesign products for document analysis, invoice operations, email workflows, and negotiations.
- Ran **daily 1:1 client meetings** to investigate production issues, understand new requirements, prioritize fixes, and turn customer feedback directly into product changes.

iOS Developer

Oct 2025 – Dec 2025

Merazine

Los Angeles, CA

- Developed a full-stack AI companion app with **real-time voice conversations and persistent memory**, along with ESP32-S3 companion hardware supporting Bluetooth/Wi-Fi setup and live two-way audio streaming.

Software Engineer

May 2025 – Aug 2025

PolicyAdvisor

Toronto, ON

- Automated insurance proposal generation with AI, cutting processing time from 5–6 hours to 2–5 minutes at 99%+ accuracy across 10+ carriers and saving **~\$100K/year**.
- Launched **viQ**, a visitor-insurance platform providing real-time quotes from 10+ carriers, coordinating a cross-functional team to ship the product in 2 weeks.
- Deployed a 24/7 AI phone assistant that answered client calls and automatically scheduled callbacks, reducing **missed leads by 40%**.

Projects

🎧 SpeakSummarize — Record. Recall 🌐 🍏 | 1K+ Downloads, \$1K MRR

- Created and launched an AI iOS app that turns recordings into structured notes using transcription, summaries, speaker separation, and personalized memory; reached **1,000+ downloads and \$1K MRR within days of launch**.
- Grew the product through Reddit and user feedback, with Android and watchOS versions in development.

🎨 Muse Sketch Studio 🌐 | Replicate AI Hackathon Winner

- Created an AI fashion-design tool that transforms a sketch and text prompt into colored concepts, model images, 3D assets, and runway videos; **won the Replicate AI Hackathon**.

Skills

Languages: Python, C/C++, TypeScript/JavaScript, Swift, SQL

Frameworks: React, React Native, Next.js, FastAPI, GraphQL

AI / ML: LLMs, RAG, AI agent evaluation, computer vision, STT/TTS

Cloud & Tools: GCP, AWS, Docker, Git, CI/CD, PostgreSQL